

C01 ASSESSMENT TOOL-1

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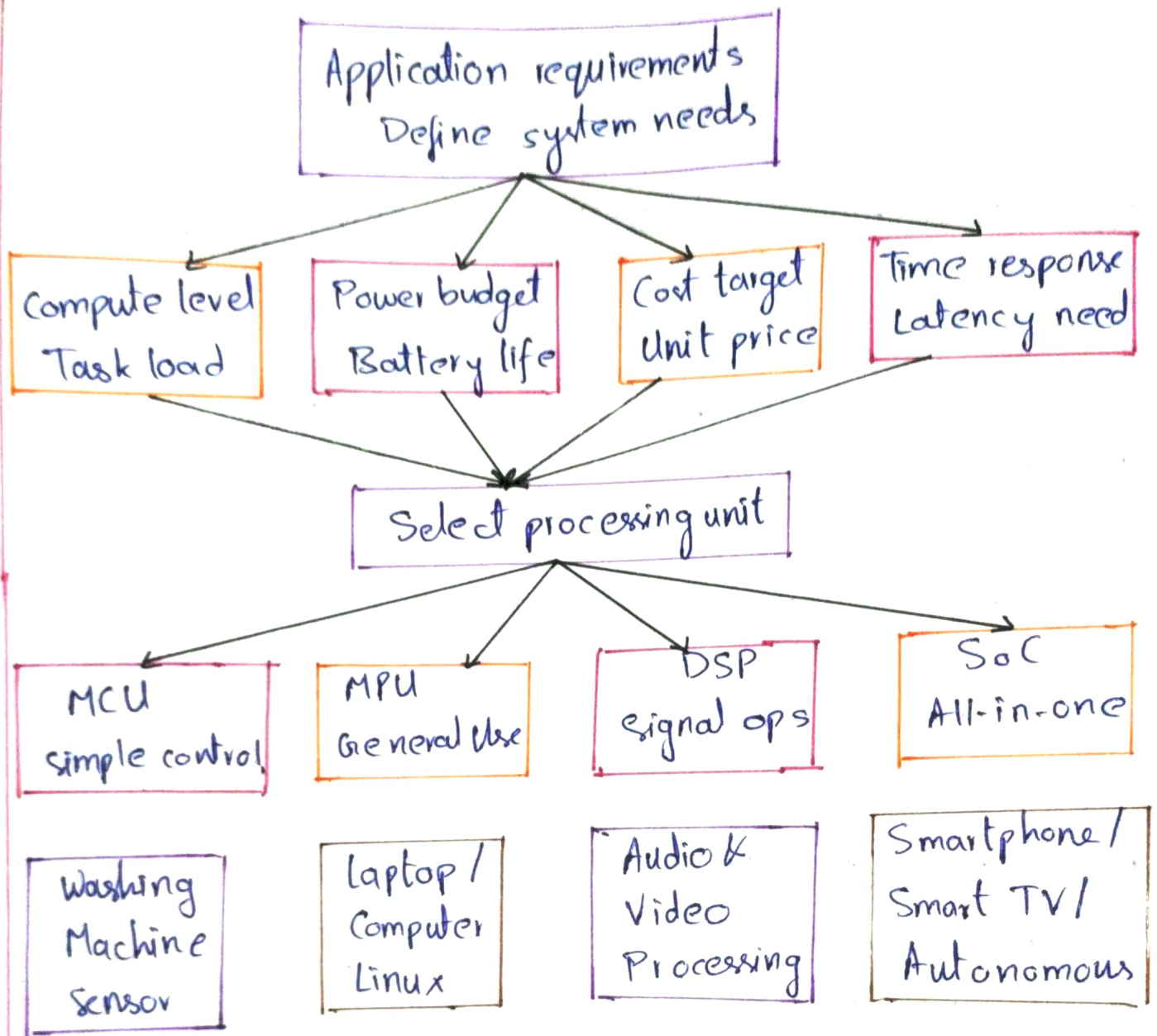
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Application Requirements and Selection of Processing Units (MCU, MPU, DSP, SoC)



Explanation of Decision Factors

1. Processing Capability

- * Refers to the speed and complexity of tasks.
- * If the application requires high computing power, choose MPU or SoC.
- * If it mainly processes audio, video, or communication signals, choose DSP.

Examples:

1. Laptop $\rightarrow$ MPU
2. Smartphone $\rightarrow$ SoC
3. Audio Processing $\rightarrow$ DSP

2. Power Constraints

- * Applications running on batteries require low power consumption.
- * MCU consumes the least power.
- * SoC provides a balance between performance and power.

Examples:

1. Smart Watch $\rightarrow$ MCU
2. IoT Sensor $\rightarrow$ MCU
3. Mobile Phone $\rightarrow$ SoC

3. Cost

- * Budget is an important factor.
- * MCU is the least expensive.
- * DSP and MPU are moderately expensive.
- * SoC is costlier because it integrates many components.

Examples:

1. Washing Machine $\rightarrow$ MCU
2. Industrial Controller $\rightarrow$ MCU
3. Smart TV $\rightarrow$ SoC

Real-Time Responsiveness

- * Some applications must respond immediately.
- * MCU and DSP provide fast real-time response.
- * SoC is used for complex real-time systems.

Examples:

1. Traffic Light Controller → MCU

2. ECG Monitor → DSP

3. Autonomous Vehicle → SoC

Real-Time Application Mapping

Application	→ Processing Unit
Temperature Sensor	→ MCU
Traffic Light Controller	→ MCU
Washing Machine	→ MCU
Laptop	→ MPU
ATM	→ MPU
Industrial Computer	→ MPU
Audio Processing	→ DSP
Speech Recognition	→ DSP
ECG Signal Processing	→ DSP
Smartphone	→ SoC
Smart TV	→ SoC
Drone	→ SoC
Autonomous Vehicle	→ SoC
IoT Gateway	→ SoC

Comparison Table

Processing Unit	Features	Real-Time Applications
MCU	Low power, low cost, built-in memory, real-time control	Washing machine, Traffic light, Sensors
MPU	High processing power, supports OS, multi-tasking	Laptop, ATM, Industrial Computer
DSP	Optimized for signal processing	Audio system, Speech recognition, ECG monitor
SoC	CPU + GPU + Memory + Peripherals on one chip	Smartphone, Smart TV, Drone, Autonomous vehicle

Choose the processing unit based on processing capability, power consumption, cost, and real-time responsiveness to achieve the best performance and efficiency for the application.